Prep Work for Ch 4

If m is an integer and $m>0$: $(1,2,3,..)$

$$b^m = \underbrace{b\cdot b\cdot b \cdots b}_{m \text{ times}}$$

$$2^3 = 2\cdot 2\cdot 2 = 8$$

If $m,n>0$ integers

$$b^m b^n = b^{m+n}$$

$$2^3\cdot 2^4 = (2\cdot2\cdot2)(2\cdot2\cdot2\cdot2)$$
$$= 2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot2 = 2^7 = 2^{3+4}$$

If $m,n>0$ integers and $b\neq 0$, $m>n$

$$\frac{b^m}{b^n} = b^{m-n}$$

$$\frac{2^4}{2^3} = \frac{\cancel{2}\cdot\cancel{2}\cdot\cancel{2}\cdot 2}{\cancel{2}\cdot\cancel{2}\cdot\cancel{2}} = 2^1 = 2^{4-3}$$

$*$) $m=n$

$$1 = \frac{2^3}{2^3} = 2^{3-3} = 2^0 \Rightarrow 2^0 = 1$$

If $b\neq 0$

$$b^0 = 1$$

$*$ $m<n$

$$\frac{2^3}{2^5} = 2^{3-5} = 2^{-2}$$

$$2^{-2} = 2^{0-2} = \frac{2^0}{2^2} = \frac{1}{2^2}$$

If $b\neq 0$, $n>0$ integer

$$b^{-n} = \frac{1}{b^n}$$

$$7^0 = 1, \quad 5^0 = 1$$
$$3^{-1} = \frac{1}{3^1} = \frac{1}{3}$$
$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$
$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$
$$\frac{1}{4^{-2}} = \frac{4^2}{1} = 4^2 = 16$$

$$\frac{7^5}{7^3} = 7^{5-3} = 7^2 = 49$$

$$(ab)^n = a^n b^n$$
$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\left(\frac{a}{b}\right)^n = \left(a\cdot\frac{1}{b}\right)^n = a^n\left(\frac{1}{b}\right)^n$$
$$= \frac{a^n}{b^n}$$

$$2^3 = 2\cdot2\cdot2 = 8, \quad 2^0 = 1, \quad 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

$$x^{\frac{1}{2}} = ?$$
$$2^{\frac{1}{2}} = ?$$
$$\left(2^{\frac{1}{2}}\right)^2 = *$$

$$(b^m)^n = ?$$

$$(2^3)^2 = (2^3)(2^3)$$
$$= (2\cdot2\cdot2)(2\cdot2\cdot2)$$
$$= 2^6 = 2^{3\cdot 2}$$

If $b\neq 0$

$$(b^m)^n = b^{m\cdot n}$$

$$* = \left(2^{\frac{1}{2}}\right)^2 = 2^{\frac{1}{2}(2)} = 2^{\frac{2}{2}} = 2^1 = 2$$

$$\left(2^{\frac{1}{2}}\right)^2 = 2$$
$$\downarrow$$
$$(\ ?\ )^2 = 2$$

$$x^2 = ?$$
$$2^2 = 4, \quad (-2)^2 = 4$$
$$3^2 = 9, \quad (-3)^2 = (-3)(-3) = 9$$
$$4^2 = 16, \quad (-4)^2 = 16$$

$$x^2 = 4$$
$$-2 \quad 2$$

$$\sqrt{4} = 2$$
$$\sqrt{9} = 3$$
$$2^{\frac{1}{2}} = \sqrt{2} \qquad 4^{\frac{1}{2}} = \sqrt{4} = 2$$
$$3^{\frac{1}{2}} = \sqrt{3}$$
$$8^{\frac{1}{3}} = \sqrt[3]{8} = 2$$

$$b^{\frac{1}{n}} = \sqrt[n]{b}$$